

AI-Based Emergency Response Agent

Simple Problem Formulation, PEAS Analysis, and Environment Classification

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1. Problem Formulation

Element	Description
Problem Statement	Unexpected emergencies like fires, medical issues, and accidents happen in places like university campuses, classrooms, and labs. Delayed response can lead to injuries, property damage, and life threats.
Agent Goal	Automatically detect emergencies, quickly figure out how bad they are (Low, Medium, or High risk), and immediately notify emergency services (Police, Fire, Medical) and response teams.
Initial State	The campus is operating normally, and no emergency or hazard has been detected yet.
Actions	Collect data from sensors/cameras, detect threats, analyze emergency types, assess risk level, send quick alerts, and guide rescue teams.
Transition Model	Normal Environment $\rightarrow$ Threat Detected $\rightarrow$ Risk Assessed $\rightarrow$ Alerts Sent $\rightarrow$ Rescue Team Deployed $\rightarrow$ Emergency Resolved & Safe Campus.
Path Cost	Minimize response time, minimize property damage, and minimize danger to human life.
Goal Test	The goal is met when an emergency is detected quickly, alerts are sent, rescue teams fix the issue, and the campus becomes safe again.

Element	Description
State Space	All possible conditions of the campus, including camera feeds, smoke levels, alert statuses, and locations of rescue teams.
Search Space	All possible steps the agent can take, from checking sensors to issuing alarms, contacting emergency services, and guiding evacuations.
Agents	Multi-agent environment. The main AI agent works together with security guards, police, fire department, and medical teams.

2. PEAS Analysis

Component	Description
Performance Measure	High accuracy in detecting real threats, low false alarms, fast response time, zero casualties, and low property loss.
Environment	University campus, including classrooms, libraries, laboratories, offices, halls, students, and staff.
Actuators	Emergency sirens, automated push notifications, warning messages, door locks/unlocks, and alert messages sent to rescue teams.
Sensors	Security cameras, smoke detectors, temperature sensors, sound detectors, and panic emergency buttons.

3. Environmental Properties

Property	Classification	Why
Observability	Partially Observable	Cameras and sensors cannot cover every single corner, and smoke or darkness can block visibility.
Determinism	Stochastic / Non-Deterministic	Emergency situations and people's reactions during a crisis are unpredictable.
Episodes	Sequential	Every action taken by the agent (like sending an early alarm) directly affects what happens next in the safety process.
Change	Dynamic	Fire spreads, crowds move, and conditions change continuously while the agent is making decisions.
Values	Continuous	Sensor inputs like temperature, continuous video feeds, and time change smoothly over time rather than in step-by-step jumps.
Agents	Multi-Agent	The AI agent operates alongside security officers, firefighters, doctors, and students.

4. Choosing the State Type

Option	Suitability	Explanation
Multi-state	Most suitable	The agent moves through 6 distinct operational stages: Normal Monitoring, Threat Detection, Severity Assessment, Alert Dispatch, Rescue Deployment, and Safety Restoration.

Final Answer: The **AI-Based Emergency Response Agent** is best classified as a **Multi-state agent** because it moves through multiple changing states while monitoring the campus, detecting threats, assessing risk levels, sending alerts, and restoring safety. It is also **Partially Observable** (sensors have blind spots) and **Stochastic** (human behavior and fires are unpredictable).

